

Hello Tuca. This is a short document that gives the expression of 2-flip local optimum probability, which involves the integration of an exponential quadratic form.

Recall that the covariance matrix is a good matrix and so is the inverse. Let $f(l)$ and $g(l)$ be the entry values of the covariance matrix C and its inverse, respectively. C and C^{-1} has dimension $\binom{N}{2}$ -by- $\binom{N}{2}$ and

$$f(0) = 4, f(1) = N - 2, f(2) = 2(N - 2). \quad (1)$$

$$\begin{aligned} g(0) &= \frac{N^2 - 9N + 16}{2(N^2 - 6N + 8)^2}, \\ g(1) &= \frac{-N^2 + 8N - 14}{4(N - 2)(N - 4)^2}, \\ g(2) &= \frac{N^4 - 14N^3 + 74N^2 - 170N + 144}{2(N^2 - 6N + 8)^2}. \end{aligned} \quad (2)$$

Then we can write

$$\begin{aligned} &\mathbb{P}(\text{2-flip local optimum}) \\ &= \frac{1}{(2\pi)^{N(N-1)/2} \det(C)^{N(N-1)/2}} \int_{[0, \infty)^{N(N-1)/2}} \exp\left(\frac{-x^T C^{-1} x}{2}\right) dx. \end{aligned} \quad (3)$$

In particular,

$$\begin{aligned} &\int_{[0, \infty)^{N(N-1)/2}} \exp\left(\frac{-x^T C^{-1} x}{2}\right) dx \\ &= \int_{[0, \infty)^{N(N-1)/2}} \exp\left(-\frac{1}{2} \left((g(2) - g(0)) \|x\|_2^2 + g(0) \|x\|_1^2 + (g(1) - g(0)) x^T M x \right)\right) dx. \end{aligned} \quad (4)$$

What we did here is that first we try to write the quadratic form in terms of L_1 and L_2 norm, then we have a remainder term. Let $\bar{S}$ and $\bar{T}$ be two index subsets with $|\bar{S}| = |\bar{T}| = 2$. The matrix M satisfies

$$M_{\bar{S}, \bar{T}} = \mathbb{1}(|\bar{S} \cap \bar{T}| = 1). \quad (5)$$